

# CosmicStride: Space-Efficient VR Exploration for LBE Venues

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(a) Screenshots of virtual reality game scenes.

(b) Gait analysis Space-Saving Effect.

**Figure 1: Overview of our demo system CosmicStride for space-efficient VR locomotion. (a) Key moments from the immersive space exploration experience. (b) Visualization of stride amplification effect: Top left – without gait-based amplification, physical and virtual distances are matched (e.g., real = virtual); Top right – with gait-based amplification, shorter physical walking yields longer virtual movements (e.g., 3m real = 5m virtual); Bottom – the system reduces required physical space.**

## Abstract

CosmicStride introduces a novel locomotion technique that enables natural walking exploration in expansive virtual environments within confined physical spaces. Leveraging vertical head motion captured via VR HMD, our gait amplification algorithm proportionally scales virtual strides to real-world steps during users’ gait cycles. This overcomes fundamental spatial constraints in Location-Based Entertainment (LBE) deployments while preserving proprioceptive continuity and reducing simulator sickness. Validated through an immersive space odyssey, the system demonstrates universal applicability in scenarios demanding authentic ambulatory exploration. Requiring no additional hardware, our approach establishes an paradigm for large-scale VR navigation.

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## 1 Introduction

Location-Based Entertainment (LBE) venues have emerged as a compelling format for immersive storytelling and interactive design. By enabling full-body movement in VR content, LBE experiences delivers a visceral sense of presence. As demand for immersive, walkable experiences rises, LBE stands at the forefront of VR’s commercial and creative expansion [Morie 2020].

In LBE experiences, natural bodily movement serves as a fundamental mode of interaction. Immersive VR storytelling significantly benefits from the physicality of walking, which enhances spatial understanding, embodiment, and presence far more effectively than artificial locomotion methods such as joystick navigation or teleportation [Langbehn et al. 2018]. However, achieving this level of immersion typically requires the physical space to closely match the scale of the virtual environment—posing a major challenge for LBE venues where real estate is costly and limited. Every additional square meter increases operational overhead. Techniques such as

Redirected Walking (RDW) [Razzaque et al. 2001] mitigate this by subtly manipulating users' movement paths to keep them within a confined area. However, the effectiveness of RDW is inherently constrained by human perceptual thresholds. Therefore, developing locomotion strategies that can convincingly extend virtual walking distances without requiring proportionate physical space—while maintaining a sense of natural movement—holds strong commercial and experiential potential for LBE deployments.

We present a locomotion technique that lets users physically walk while virtually covering greater distances, with no extra hardware. Using only standard head-mounted display (HMD) sensors [Angelov et al. 2020], our system detects the user's natural walking rhythm in real time and synchronizes subtle adjustments to users' virtual movement with specific moments in their gait. As a result, each physical step translates into a longer virtual stride, enabling traversal of larger virtual environments while preserving the illusion of natural walking like Figure 1 (b).

In our demo, CosmicStride, users can embark on a thrilling journey as astronauts, experiencing rocket launches, space crises, and an unforgettable Mars exploration mission. This technique dramatically reduces the required physical space for VR experiences, offering a new and way to navigate vast virtual worlds in a compact real-world footprint.

## 2 Scenario Design and User Journey

Participants begin at Satellite Launch Center, surrounded by cheering crowds witnessing a rocket launch, immersing users in the historic moment. Abruptly, the scene shifts to a restored mission control room, participant pressed a button, triggering an emergency temporal jump that teleport them into the space capsule.

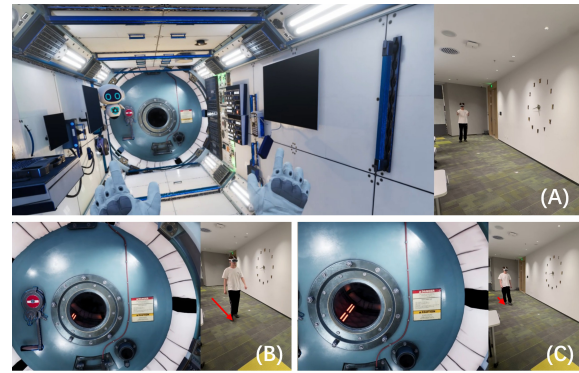
After the space capsule completes the docking with the space station. Participants watched the history of space exploration through the screens inside the space station: Orbits of artificial satellites, Apollo landings, and Human space missions. Without warning, meteoroid impacts rupture the station hull, triggering rapid pressure loss warnings. Robot NPC remotely opens the airlock, compelling users to conduct an urgent repair: replace damaged solar array components and using tools to seal hull breaches.

Subsequently, the participants received the mission to Mars and once again entered the space-time rift. This time, the participants saw the magnificent landforms of Mars, and they could deeply experience human resilience and interstellar exploration.

## 3 Gait-Aware Stride Amplification

Our system detects users' gait phases in real time using only inertial sensors embedded in standard HMDs, requiring no external hardware. We rely on the vertical component of head motion, as vertical acceleration during walking exhibits clear periodic patterns aligned with gait cycles.

We derive vertical velocity by differentiating the HMD's positional tracking data sampled at 72Hz, then apply double exponential smoothing to reduce noise while preserving temporal dynamics. We then approximate vertical acceleration by differencing consecutive smoothed velocity samples. Zero-crossings in this signal reveal rhythmic fluctuations aligned with gait, which we segment into alternating phases of stability and balance.



**Figure 2: (A) Range of the space station (B) Walk in proportion to the actual space (C) Enable gait analysis to reduce actual space walking**

Tracking these markers in real time produces a robust temporal signal representing the user's natural walking rhythm, tolerant to typical sensor noise and small head movements such as looking around. Based on this rhythm, the system applies subtle, phase-aligned adjustments to the user's virtual movement speed during each step. As a result, each physical step is imperceptibly translated into a longer virtual stride, enabling traversal of expansive virtual environments within confined physical spaces while preserving the illusion of natural walking. All processing runs in real time on standard VR hardware, ensuring low latency and straightforward deployment without added cost or complexity.

## 4 Summary

This work presents gait analysis that enables expansive virtual exploration within confined physical spaces through gait-phase detection using standard HMD sensors. By leveraging vertical head motion dynamics during natural walking, our technique imperceptibly scales virtual displacements in sync with users' rhythmic gait patterns—preserving proprioceptive continuity while overcoming spatial constraints inherent to Location-Based Entertainment deployments. Demonstrated through an immersive space odyssey, the system validates its efficacy for LBE applications without specialized hardware.

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